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ORIGINAL ARTICLE

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The bilateral leg strength deficit is present in old, young and adolescent females during isokinetic knee extension and flexion

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Abstract The bilateral limb deficit (BLD) describes the difference in maximal or near maximal force generating capacity of muscles when they are contracted alone or in combination with the contralateral muscles. A deficit occurs when the summed unilateral force is greater than the bilateral force. This study examined the presence of the BLD during isokinetic knee extensions and flexions in a group of adolescent females (n = 8, mean of 15 ± 1 years) and compared with previously reported data by this researcher of adult and older females. Data were collected from subjects during slow (45 deg/s) isokinetic knee extensions and flexions and it was found that a BLD exists during both extension and flexion regardless of age. Furthermore, this study is the first to examine the presence of the deficit in an adolescent population. Myoelectric signal (MES) data showed that there is no difference between bilateral and unilateral isokinetic knee extensions and flexions regardless of age group.

Keywords Bilateral limb deficit · Isokinetic · Age · Torque · Gender

Introduction

The bilateral limb deficit (BLD) describes the difference in force generating capacity of muscles when they are contracted alone, or together with the contralateral muscle group. A deficit occurs when the summed unilateral force is greater than the bilateral force. A BLD exists for both large and small muscle groups in a variety of movement patterns, in both males and females and in athletic and nonathletic individuals, and also in subjects with motor disorders (Archontides and Fazey 1993; Henry and Smith 1961; Howard and Enoka 1991; Jakobi and Chilibeck 2001; Ohtsuki 1983, 1994). Also, BLD is reported in older adults (Owings and Grabiner 1998; Hernandez et al. 2003). The BLD may reflect neural inhibition during bilateral contractions (Archontides and Fazey 1993; Howard and Enoka 1991), but that is not established (Archontides and Fazey 1993; Jakobi and Chilibeck 2001).

Some studies have suggested that the BLD may be related to a reduced capacity to recruit fast-twitch fibers (Kawakami et al. 1998, Vandervoort et al. 1984). If this was the case, one might expect that older subjects with fewer fast-twitch fibers would demonstrate less of a BLD. However, studies on BLD with respect to age have been contradictory. The BLD is demonstrated in older adults during slowly developing ramp and hold contractions of a similar magnitude to that found in younger adults (Owings and Grabiner 1998). Similarly, there is no difference in BLD either with respect to maximal or submaximal contractions during isometric elbow flexion from older and younger subjects (Hernandez et al. 2003). Also, Häkkinen et al. (1996a, 1996b, 1997) found that while there are differences in overall muscle strength between males and females and between people from 30 to 70 years of age, there is a little difference between bilateral and unilateral strength in rapid and step contractions. Recent evidence (Kuruganti et al. 2005) has shown that the BLD is present in similar amounts in both young adults (18–35 years old) and older (55–75 years old) adults. There has been little work that has investigated the presence of the BLD in younger subjects such as adolescents. The present study also extended the author's previous work (Kuruganti et al. 2005) and compared the BLD from three different age groups of females to determine if the BLD is influenced by physiological changes due to aging. It was hypothesized that the BLD would not be affected by age.

The myoelectric signal (MES) measures the result of the neural commands sent to the muscle and would

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provide evidence of the neural mechanisms responsible for the deficit. The amplitude of the MES depends on the membrane properties of the muscle fibers as well as the timing of the motor unit action potentials and therefore reflects both peripheral and central properties of the neuromuscular system (Farina et al. 2004). Studies to date have been equivocal regarding the relationship between the force and MES deficits. Previous studies of the BLD and the MES have been inconclusive and in many cases MES data have not paralleled force or torque data under the same conditions. Some studies have shown that the amplitude of the MES is lower under bilateral conditions versus unilateral conditions (Kawakami et al. 1998; Koh et al. 1993; Oda and Moritani 1994; Ohtsuki 1981, 1983; Rube and Secher 1990; Steger and Denoth 1996; Van Soest et al. 1985; Vandervoort et al. 1984). However, several other studies have shown that there is no statistically significant difference (Howard and Enoka 1991; Owings and Grabiner 1998; Schantz et al. 1989). It has been suggested that a decline in MES, in parallel with force deficit during bilateral contractions would imply a neural basis to the deficit (Howard and Enoka 1991). It was hypothesized that the MES would decline during bilateral contractions correspondingly with force production.

Methods

Eight adolescent females participated in the strength testing and their data were compared to previous data of adult and old females collected by this researcher (Kuruganti et al. 2005). The old group ($n = 7$) had a mean ($\pm$ SD) age of 63 $\pm$ 6 years, height of 162 $\pm$ 6 cm and weight of 70 $\pm$ 18 kg. The adult group ($n = 8$) had an age of 31 $\pm$ 7 years, a height of 166 $\pm$ 6 cm and a weight of 72 $\pm$ 12 kg. The young group ($n = 8$) had a mean of 15 $\pm$ 1 years, height = 167 $\pm$ 5 cm and a weight of 58 $\pm$ 6 kg.

The subjects were required to visit the laboratory twice, once for familiarization and once for the strength testing as approved by the University of New Brunswick Research Ethics Board. Subjects signed an assent form with parent consent.

Familiarization

Prior to acceptance to the program, the subjects were asked to attend a familiarization session consisting of a screening process to ensure that they were healthy and fit. The subjects were provided with an overview of the testing session and asked to read and sign an informed consent form and a screening questionnaire including a physical activity questionnaire. Subjects aged 55–75 years were also encouraged to discuss their participation in the program with their family doctor.

Instrumentation

A Cybex II (Lumex Incorporated, NY, USA) isokinetic dynamometer was used to conduct isokinetic knee extensions and flexions. An eight-channel bioelectric signal data acquisition system recorded torque and position using an acquisition system amplifiers (Tektronix, TM 5006, Lake Mary, FL, USA). The gains of the amplifiers were set at 1 and the filters at 10 Hz. The data were sampled at 1,000 Hz and analog to digital converted using a custom-built data acquisition software (ACQA, V. 3.1). Further processing was performed using Matlab (Version 5.3.1, The MathWorks Inc., Natick, MA, USA).

Strength testing

Prior to each of the testing sessions the subjects completed 5 min of brisk walking on the treadmill to warm up their muscles. During the isokinetic extensions and flexions the subjects were seated in the dynamometer with their ankles firmly strapped to the extension bar. In order to prevent any interference from the upper body, a five-point harness was used to keep the subjects from moving during the contractions and they were instructed to look straight ahead. The ankles were strapped to the input adapter of the dynamometer and the subject extended the leg upwards and flexed the leg back towards the chair. The subjects were asked to apply as much effort as possible during slow (45 deg/s) bilateral isokinetic knee extensions and flexions, while provided visual feedback by an oscilloscope. No break was provided between contractions.

During each of the sessions the subjects were required to complete two maximum voluntary contractions for three conditions, bilateral (BL), unilateral right (ULR) and unilateral left (ULL). Subjects began with the bilateral conditions, followed by the unilateral contractions randomly presented. For each of the three conditions, the trial in which the greatest torque was produced was used for data processing. Two-minute breaks were provided between contractions to limit muscle fatigue and subjects were provided with 3–5 submaximal familiarisation contractions prior to data collection for each protocol (BL, ULR, ULL).

A general linear model (GLM) analysis of variance (ANOVA) was conducted on the strength performance for the three groups. When a group main effect was detected, a post-hoc (Tukey) analysis was run to determine which of the three groups were different from each other.

Bilateral limb torque

The BLD was quantified similar to previous studies (Howard and Enoka 1991; Jakobi and Cafarelli 1998; Jakobi and Chilibeck 2001; Ohtsuki 1983). The bilateral

leg ratio (BLR_{torque}) was calculated for the extension and flexion phases using the peak torque data from the dynamometer as: BLR_{torque} (%) = (Peak torque BLR/(Peak torque ULL + Peak torque ULR)) × 100 (Ohtsuki 1983).

A GLM ANOVA was run for each of BLR_{torque} for extension and BLR_{torque} for flexion to determine if the BLR_{torque} values differed across the three age groups. A BLR less than and significantly different from 100% was considered a deficit. A Student's *t*-test was used to determine if the BLR was significantly different from 100%.

Myoelectric signal data collection

For the MES data, the signals were fed through a set of pre-amplifiers (with gains of 600), and then were optically isolated, amplified and band limited to 0.1 Hz–1 kHz. Anti-aliasing filters were set at 500 Hz. The data were then converted from analog to digital using a PC with custom-built data acquisition software (ACQA, V. 3.1) under the LabVIEW (National Instruments) environment. Both the torque and MES data were sampled at 1,000 samples/s. Further processing was performed using the Matlab (Version 5.3.1) programming language.

The number of channels of MES collected was limited by the instrumentation to four, therefore one muscle from each of the right and left leg quadriceps and hamstrings were chosen. The four knee extensor and flexor muscles were right/left vastus lateralis (VL) and right/left biceps femoris (BF). Bipolar surface electrodes (Duotrode silver–silver chloride electrodes, Myo-tronics, Inc., interelectrode spacing = 21.0 ± 1 mm) were placed over the muscle belly of the VL at 2/3 on the line from the anterior spina iliaca superior to the lateral side of the patella (SENIAM.org). A second bipolar electrode pair was placed over the midline of the BF at a position one-half the distance between the ischial tuberosity and the popliteal fossa. A common reference electrode was placed over the crest of the tibia.

In order to ensure a good electrode–skin interface, prior to the application of the electrodes, the subject's skin was first abraded and then cleaned with alcohol. The surface electrodes were then placed over the belly of each muscle. The electrodes were not moved during the entire testing session.

While the subjects were seated, their legs were in direct contact with the seat of the dynamometer. In order to limit any motion artifact at the electrodes placed over the hamstrings, doughnut shaped foam-padding was placed around each electrode and on the subject's legs. This ensured that the electrodes did not come into contact with the seat cushion during the contractions.

The amplitude of the MES was quantified using a Root Mean Square (RMS) calculation. A T-second window of MES data, centred at the time at which peak torque occurred, was used to estimate muscle activity. The amplitude of the MES was estimated with a RMS calculation as described by Eq. 1.

$$A(T_p) \equiv \mathrm{RMS}\{m(t)\} = \sqrt{\left(\frac{1}{T}\int_{T_p-T/2}^{T_p+T/2} m^2(t)\mathrm{d}t\right)}, \qquad (1)$$

where $A(T_p)$ (mV) = RMS value of the MES from a T-second window centred at time T_p

$m(t)$ = raw MES signal at time t
T_p = time of peak torque
T = window length (0.5 s in this experiment)

Each of the extension and flexion phases took approximately 2 s to complete (total of 4 s from start of contraction to completion). During the entire contraction the subject was motivated to "push as hard" as they could and produced a maximal voluntary contraction throughout the entire 4 s. Therefore, a 0.5-s window centred at peak torque was determined to be sufficient to estimate the muscle activity as a measure of the peak torque.

A GLM analysis of variance (ANOVA) was run for each of the RMS values obtained for each contraction to determine if the mean amplitude of the MES differed across the three age groups. A paired *t*-test was also used to determine if there were differences in the MES between bilateral and unilateral contractions. The alpha level was set to 0.05.

Results

Unilateral and bilateral strength

The mean unilateral and bilateral strength are shown in Table 1. There was a difference in bilateral extension (P = 0.04), unilateral left extension (P = 0.007), bilateral flexion (P = 0.004), and unilateral left flexion (0.004) torques due to age. Generally, younger females displayed greater strength than the adult and older adult females (Table 1).

BLR_{torque}

A significant bilateral deficit was detected in all subjects during extension and flexion (Table 2). There was no difference in the BLR due to age for either contraction (P = 0.175 for extension and P = 0.249 for flexion)

To determine if the BLR was linearly related to the amount of bilateral strength, a correlation analysis was conducted. It was found that the BLR_{torque} was correlated with the bilateral torque. For extension (across all three groups), BLR_{torque} was reasonably correlated with bilateral extension torque (r = 0.458, P = 0.028) and for flexion the BLR was correlated with bilateral flexion torque (r = 0.474, P = 0.022).

Table 1 Bilateral and unilateral torque for extension and flexion across three age groups

Contraction	Torque (Nm)		
	Younger	Adult	Older
Bilateral extension	171.9 ± 61.5	142.9 ± 24.4†	109.2 ± 36.8*
Unilateral right extension	111.8 ± 33.6	94.4 ± 19.4	80.4 ± 28.3
Unilateral left extension	118.1 ± 26.6	86.1 ± 16.8†	77.8 ± 25.9*, ††
Bilateral flexion	107.9 ± 34.5	70.6 ± 11.0†	65.5 ± 19.4*, ††
Unilateral right flexion	67.6 ± 18.9	51.2 ± 12.0	48.1 ± 17.1
Unilateral left flexion	72.4 ± 16.6	48.5 ± 11.2	49.5 ± 13.1*, ††

Note that the torque differs across age group for bilateral and unilateral left extension torque and bilateral and unilateral left flexion torque
Values are mean ± SD
†Post-hoc (Tukey) analysis indicates significant difference between adult and young groups ($P < 0.05$)
††Post-hoc analysis indicates significant differencesold and young groups ($P < 0.05$)
†Unilateral left extension ($P = 0.0325$)
††Unilateral left extension torque ($P = 0.0089$)
†Bilateral flexion torque ($P = 0.0142$)
††Bilateral flexion torque ($P = 0.0071$)
†Bilateral extension torque ($P = 0.0316$)
††Unilateral left flexion torque ($P = 0.0122$)

Myoelectric signal data

The MES data collected during unilateral and bilateral trials were compared to determine if there were differences in the MES amplitude depending on one or two legs contracting. No significant differences were detected between the bilateral and unilateral MES amplitude across the three age groups for any of the contractions.

Discussion

Previous work (Kuruganti et al. 2005) showed that the BLR exists in both adults and older adults during a dynamic knee extension and flexion. This study furthers these results by examining a younger, adolescent population. No similar study examined the presence of the bilateral deficit in isokinetic knee extensions and flexions in an adolescent population.

It was found that the BLD was present during both knee extensions and flexions in all three age groups of females. Interestingly, while the strength values differed across the three groups, there was no detectable difference in the BLR_{torque} across the three groups. Previous studies of older adults did not detect a BLD (Häkkinen et al. 1996a, 1996b, 1997) or found BLD in older people, comparable with results for younger people, without testing a younger group (Owings and Grabiner 1998). The findings from the older adult group presented here are similar to those of Owings and Grabiner (1998). The results from the younger group show that the deficit is present in young, adolescents similar to adults and older adults.

Lower limb dynamic knee extension movements show a range of BLR_{torque} from 78 to 108% (Owings and Grabiner 1998; Taniguchi 1997; Vandervoort et al. 1984; Coyle et al. 1981; Brown et al. 1994) and may be dependent on experimental protocols used, the range of the contraction velocity (slow to fast contractions), and the fitness of the participants.

Only two previous studies have examined knee flexion (Brown et al. 1994, Kuruganti et al. 2005) and both found similar BLR when compared to knee extension. This study found that the mean BLR_{torque} across three groups of females was similar for flexion (72%) and extension (74%).

The data from all the three groups showed that there was no difference in the MES amplitude between bilateral and unilateral contractions for the extension or flexion. This suggests that the deficit may not be due to a limitation in neural mechanisms. Previous studies of the BLD and the MES have been inconclusive and in many cases MES data have not paralleled force or torque data under the same conditions. Some studies have shown that the amplitude of the MES is lower under bilateral conditions versus unilateral conditions (Kawakami et al. 1998; Koh et al. 1993; Oda and Moritani 1994; Ohtsuki 1981, 1983; Rube and Secher 1990; Steger and Denoth 1996; Van Soest et al. 1985; Vandervoort et al. 1984). The data found in this study agree with the other studies that have shown that there is no statistically significant difference (Howard and Enoka 1991; Owings and Grabiner 1998; Schantz et al. 1989). This suggests that the deficit is not solely the result of reduced motor unit activation.

This study found that the bilateral leg strength deficit is present during isokinetic knee extensions and flexions in adolescent females and is comparable to adult and older females. Finally, there was no difference detected in the MES amplitude between bilateral and unilateral knee extension and knee flexion suggesting that the deficit is not simply due to reduced motor unit activation.

Table 2 Bilateral leg ratio torque (BLR_{torque}) for extension and flexion across three age groups

Group	Contraction	Bilateral limb ratio (%)
All subjects	Extension	74.6 ± 9.8
All subjects	Flexion	72.2 ± 9.8
Adult females	Extension	79.5 ± 4.5
	Flexion	72.3 ± 10.3
Old females	Extension	70.2 ± 12.6
	Flexion	67.6 ± 7.7
Young females	Extension	73.6 ± 9.8
	Flexion	76.2 ± 10.3

Note that the mean BLR_{torque} is not statistically significant across the three groups suggesting that the deficit is similar regardless of age
Values are mean ± SD

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